

Varun Hariharan

varun.hariharan203@gmail.com | (206) 519-4047 | [LinkedIn](#) | [Portfolio](#) | [GitHub](#)

EDUCATION

Northeastern University

Master of Science in Artificial Intelligence

Sep 2026 – May 2028

University of Washington

Bachelor of Science in Human Centered Design & Engineering; Bachelor of Science in Economics

Sep 2022 – Jun 2026

PROJECTS

Monte Carlo Portfolio Risk Simulator (Python, NumPy, Streamlit, Claude API)

Apr 2025 – Present

- Built an AI-powered portfolio risk simulator that converts plain-English portfolio descriptions into multi-asset Monte Carlo simulations and generates natural-language explanations of portfolio risk.
- Modeled thousands of correlated market scenarios (up to 10,000 simulated paths) using Geometric Brownian Motion, historical covariance matrices, and Cholesky decomposition across 5+ years of real market data; calculated VaR, CVaR, percentile ranges, and probability of loss.
- Developed a rolling out-of-sample VaR backtesting framework across 1,000+ trading days using historical market data and the Kupiec proportion-of-failures test, confirming a 4.60% breach rate against a 5% target ($p=0.56$) under normal conditions.

AI Data Analyst (Python, FastAPI, Streamlit, Claude API)

Jun 2026 – Present

- Built an AI-powered conversational data analyst that allows users to upload CSV datasets and perform data analysis through natural-language queries.
- Integrated an LLM-powered code generation pipeline that translates user questions into executable Python/Pandas analysis and dynamically processes uploaded datasets.
- Developed a FastAPI backend and React frontend for dataset ingestion, AI interaction, code execution, and presentation of analytical results.
- Implemented automated visualization and natural-language insight generation, enabling users to receive charts, statistical results, and explanations without writing code.

Research Clipper (Python, FastAPI, PostgreSQL, pgvector, Claude API)

Jul 2026 – Present

- Built a browser-based research tool with a UW collaborator that lets users clip web content and query saved sources through a citation-grounded RAG pipeline, generating answers with claim-level source attribution and declining unsupported queries.
- Designed and evaluated the retrieval and grounding pipeline across 16 direct-recall, negative-case, and cross-source synthesis questions, achieving 16/16 correct responses, including 3/3 unsupported queries correctly declined.

EXPERIENCE

CoMotion Labs — Patent Quest (Undergraduate Capstone) (Figma, Prototyping, Playtesting)

Jan 2026 – Jun 2026

UI/UX Designer

- Made the core product decision to translate abstract IP/patent strategy concepts into gameplay mechanics, prioritizing intuitive understanding over strict accuracy to reach early-stage innovators
- Used playtesting data to identify which decision-flow friction points to fix first, informing a prioritized iteration roadmap

Valorant Registered Student Organization (Figma, Accessibility Design, Usability Testing)

Jan 2025 – Apr 2025

UI/UX Designer

- Identified accessibility and navigation as the highest-leverage problems for a 1.5K+ member organization, based on synthesized findings from usability testing with 15+ users
- Prioritized 4+ rounds of iteration based on direct evidence of where users struggled, not aesthetic preference

PRODUCT & RESEARCH

Seesaw (Sep 2024 – Mar 2025): Goal-tracking app for UW students; research with 25 participants, 8-user usability testing

City (Sep 2024 – Dec 2024): Digital festival experience; branding, competitor analysis, mobile app concept

SKILLS

AI/ML: Claude API, RAG, Tool Use / Function Calling, Prompt Engineering, Sentence Embeddings, pgvector, Vector DB, PyTorch

Data & Analysis: NumPy, Pandas, Matplotlib

Programming: Python, R, Java, JavaScript, HTML/CSS, React, PostgreSQL

Tools: FastAPI, Streamlit, Git, Figma, VS Code, Vercel